

DAY 2 TCP/IP SUITE THEORY

Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

ANS:

Here is how to identify one packet for each layer while refreshing Flipkart in Wireshark:

1. **Application Layer:** Look for **HTTP/2**, **HTTP/3**, or **TLS** (showing requests for Flipkart domain paths and handshakes).
2. **Transport Layer:** Look for a **TCP** (or **UDP**) packet; expand its header to see *Source/Destination Ports* (e.g., port 443) and sequence numbers.
3. **Internet Layer:** Expand the **IPv4** or **IPv6** header to find logical *Source and Destination IP addresses* between your device and Flipkart's server.
4. **Network Access Layer:** Expand the bottom **Ethernet II** or **802.11** header to view physical *Source and Destination MAC addresses*.

Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

ANS:

Here is a quick side-by-side comparison of the OSI and TCP/IP models with real-world app examples:

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Real-World App Example (WhatsApp / Zomato)
7. Application	4. Application	WhatsApp: Typing and sending a message ("Hey!").
6. Presentation	<i>(Included in Application)</i>	WhatsApp: Encrypting and formatting text/emojis.
5. Session	<i>(Included in Application)</i>	Zomato: Maintaining an active ordering session.

4. Transport	3. Transport	WhatsApp: TCP ensuring reliable, ordered delivery.
3. Network	2. Internet	Zomato: Routing IP packets across the internet.
2. Data Link	1. Network Access	WhatsApp: Moving frames across local Wi-Fi via MAC addresses.
1. Physical	<i>(Included in Network Access)</i>	Zomato: Wi-Fi radio waves transmitting binary bits through the air.

Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.

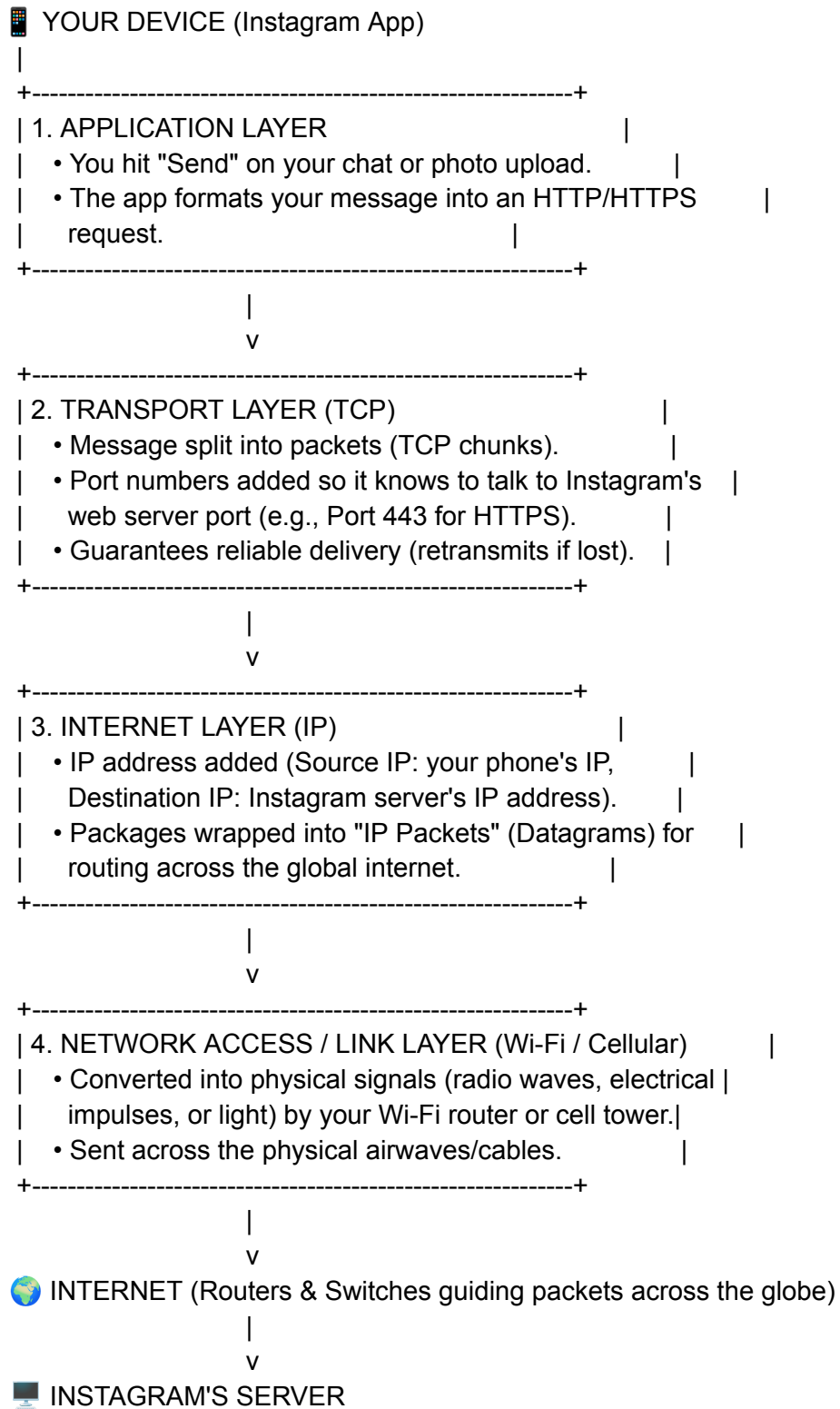
ANS:

Here are the core TCP header fields and their functions:

1. **Source Port:** Identifies the sending app's port so the server knows where to reply.
2. **Destination Port:** Specifies the target app's port on the server (e.g., port 443).
3. **Sequence Number:** Tracks data order so the receiver can reassemble segments correctly.
4. **Acknowledgment Number:** Confirms received data and indicates the next expected byte.
5. **Flags:** Control connection states (e.g., **SYN**, **ACK**, **FIN** for handshakes and teardowns).
6. **Window Size:** Manages flow control by telling the sender how much data can be accepted.
7. **Checksum:** Performs error-checking to detect corrupted header or data packets.

Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

ANS:



Summary of What Happens at Each Layer:

1. **Application Layer:** Prepares the readable message or media file into a web-ready format using protocols like HTTPS.
2. **Transport Layer (TCP):** Breaks the heavy message down into manageable sequence-numbered packets and ensures error-free delivery.
3. **Internet Layer (IP):** Stamps the packets with sender and receiver IP addresses so routers across the world know exactly where to forward them.
4. **Network Access / Link Layer:** Translates the digital data into physical radio waves (Wi-Fi/4G/5G) to physically exit your device.

Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

ANS:

Here is an analysis of a representative packet from a Wireshark capture:

- **Packet Identification:** A TCP [SYN] packet.
- **TCP/IP Layer: Transport Layer** (Layer 3 of the TCP/IP model / Layer 4 of the OSI model).
- **How It Was Identified:**
 - Looking at the **Protocol** column in Wireshark showed TCP.
 - Expanding the **Transmission Control Protocol** header in the packet details pane revealed source and destination ports (e.g., source port 51234 to destination port 443), sequence numbers set to zero, and the SYN flag checked to initiate a connection.
- **Real-World App Action:** This represents the exact moment you tap to load **BookMyShow** to book movie tickets, or open **WhatsApp** to chat—your phone sends this initial TCP handshake packet to establish a secure connection with the app's server before any ticket data or messages can be sent.